# Beyond Division of Labor: How Roles Shape Feedback and Recovery in Multi-Agent Systems

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## Abstract

Multi-agent large language model (LLM) systems commonly assign distinct roles such as Planner, Executor, and Reviewer. In this paper, we investigate what these roles actually change beyond their intended division of labor. We evaluate three open-weight models on the Omni-MATH benchmark and two of them on the SciBench benchmark. Throughout these experiments, we hold the agents, agent topology, call budget, and ground-truth evaluator constant. We find that deleting all role instructions largely eliminates functional specialization, yet the team's first-pass accuracy does not drop. Roles also allocate decision rights. A role-structured Reviewer often withholds approval and forces internal deliberation. Consequently, its team submits answers to the external evaluator less frequently and receives less corrective feedback. In an exploratory comparison where feedback is accessible to both teams, the role-free team recovers from a failed first answer more often across all five model-benchmark settings and achieves higher final accuracy. A post hoc analysis reveals that this role-free advantage disappears when comparing the two teams at an equal number of evaluator verdicts. In a second exploratory intervention, run for the gemma-4 model on 400 Omni-MATH problems, we preserve the role prompts but force a submission in every cycle (one pass of the three agents), and the role-structured team's recovery rate then matches that of the role-free team. We conclude that roles govern decision rights and information flow as well as work allocation. Evaluating multi-agent organizations therefore requires looking beyond role fidelity to analyze decision policies, feedback access, and failure recovery.

## Keywords

Multi-agent systems, large language models, role specialization, division of labor, organizational structure, ablation

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## 1 Introduction

Researchers usually present a multi-agent large language model system as a detailed diagram. It features components like a Planner that decomposes without solving, an Executor that carries out the plan, a Reviewer that critiques rather than re-derives (PER), and arrows for who hands what to whom [21, 37, 43]. This diagram is read as a mechanism that explains where the system's competence comes from and justifies the cost of multiple agents. In practice, however, what runs is simply a set of prompts handed to a language model. The model may honor the intended division of labor, ignore it or satisfy its surface constraints while solving the entire problem in every box. A simple accuracy score cannot distinguish between these scenarios. A pipeline whose agents have quietly collapsed into identical solvers returns the same final score as an organization that works exactly as drawn.

Beyond merely dividing work, roles also dictate critical decisions. In a PER team, the Reviewer does not just check the answer. It also decides when the candidate is ready for submission to an external evaluator. This decision is crucial because the evaluator provides corrective information, namely a verdict and a repair hint, which internal deliberation cannot generate. A conservative Reviewer saves evaluations. This behavior is practical when evaluations are expensive. However, every withheld submission is a missed opportunity to discover that an answer is incorrect. Therefore, we investigate what roles alter in a multi-agent system beyond the intended division of labor. We specifically examine how roles govern the team's access to external corrective information and how this access impacts recovery from failure.

To isolate the effect of roles, we fix the agent topology, the actor model (the language model that plays all three agents), the call budget, and the ground-truth evaluator. We then vary the organization in two directions. First, we completely delete every role instruction. Second, we strictly enforce the division of labor at the interface using typed message formats. Because feedback typically reaches each agent through its prompt, a role ablation also rewrites the feedback channel. We therefore carefully record the exact information each condition receives (§4). In our main comparison, both teams have equal access to the evaluator's verdict and hint. While access is matched, the realized number and content of verdicts differ. This difference is what roles control.

Deleting roles largely dismantles the intended division of labor (aided by the fact that the role-free prompt asks each agent for its current answer). However, consistent with prior role ablations [10, 35], the team's first-pass accuracy does not decline. The crucial difference emerges only after a rejected first answer. In an exploratory comparison where feedback is available to both teams, the role-free team successfully recovers more of its failed trajectories (runs on a problem whose first answer was rejected) across all five actor and domain settings. This leads to a higher final accuracy. The evidence points to the Reviewer's decision rights. A role-structured Reviewer frequently withholds approval, often even before any feedback exists. Consequently, the role-structured team submits less often and receives fewer verdicts. Under a capped budget, a withheld submission is a lost chance to recover. A post hoc comparison at equal numbers of evaluator verdicts removes the



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role-free advantage. Furthermore, an exploratory intervention for the gemma-4 model on 400 problems keeps the role prompts intact but forces the team to submit an answer in every cycle (one pass of the three agents). This intervention raises the role-structured team's recovery to the role-free team's level on the same problems. Enforcing the division of labor at the interface also loosens this Reviewer gate and raises recovery, though resolvably only for the Llama-3.3 model. However, 8% to 17% of enforced trajectories never finish, because malformed or boundary-violating outputs use up the allowed regenerations, so end-to-end accuracy does not improve. Ultimately, roles shape two distinct factors simultaneously. They determine who does which work, and they dictate when the team asks for correction. These two factors affect system performance in different ways.

We summarize our contributions as follows:

(1) We introduce a budget-matched design that varies organization in two directions and strictly documents the information each condition receives. This includes a feedback-withheld diagnostic that shows what a role ablation measures when it also removes the feedback channel.
(2) We provide empirical evidence across three actors and two domains that roles successfully induce functional specialization without improving first-pass accuracy.
(3) We provide evidence that the Reviewer's approval governs the team's feedback opportunities. We support this claim using submission logs, a post hoc verdict-matched comparison, and an exploratory intervention that removes the submission gate while keeping role prompts intact.
(4) We outline implications for evaluating multi-agent organizations. We recommend reporting recovery rates after a failed first attempt and treating decision rights and information pathways as core components of the organizational structure (§6).

## 2 Related Work

*Roles in organizational models.* Organization-centered multi-agent systems give roles a formal existence independent of their occupants. Moise$^+$ separates structural, functional and *deontic* specifications [24]. OperA makes role adoption an explicit social contract [7], and electronic institutions specify norms outside the agents [13], to be regimented (violation made impossible) or sanctioned [19]. An LLM role prompt is the degenerate case: functional and deontic clauses in prose, with nothing to make them bind. Our role-free condition keeps the positions, the communication graph and the turn order, but deletes these clauses. Conversely, our enforced condition regiments the functional part at the interface. In this paper, adaptability means revising an answer across attempts on the same problem, not reorganizing the team [22]. Teamwork models make repairing a failed joint plan a team obligation [41], which a prompt can state but not secure.

*Role specialization and role drift.* Prompted roles are known to be unreliable. Agents disobey the role specification across annotated traces [4], and *role drift* has a training-time account [3]. Role-label conditioning fails to induce the intended specialization [35], and agents collapse into highly similar rationales [36]. Differentiation can even arise *without* assigned roles [12]. Recent enforcement-side work makes role scope part of the system structure rather than advice in the prompt [15, 40]. While role drift and the decoupling of role quality from task performance are known [3, 44], we measure them under a controlled ablation.

*Redundancy and parity.* Dylan et al. [10] hold total LLM calls fixed and find an evolved Planner, Executor and Critic team no better than an evolved single agent. Zhang et al. [47] find that many reported multi-agent efficiency gains arise where one agent suffices. This objection is why we report a single-agent reference (§4). On the same premise, Chen et al. [6] prune redundant agents.

*Feedback, revision and answer lock-in.* Iterative refinement with self-generated or tool feedback [17, 30, 38] and multi-agent debate [9, 29] are standard patterns. Models often repeat earlier outputs under exact verifier feedback [25]. Under self-reflection, this failure is called Degeneration-of-Thought [29], and belief entrenchment is studied directly [33]. Huang et al. [23] report that self-correction degrades accuracy only in the *intrinsic* setting, where the model judges its own answer, and improves it when the correct label serves as a stopping rule [see also 27, 39]. Our loop is gated by such an oracle (an evaluator that knows the correct answer), where more attempts buy more coverage, the share of problems solved by at least one attempt [1]. This is the regime in which the role-structured team's restraint costs it. Closest to our lever, Chen et al. [5] vary only the *role label* on a byte-identical erroneous claim and move explicit-correction rates by 23 to 93 percentage points. However, they do so in a single agent without budget matching.

*Coordination versus capability.* Canedo [2] frames structured specification as a *capability equalizer* whose value falls with model strength. Theoretical [46] and complementary [8] accounts agree. Our enforced condition reproduces that direction only under a scoring rule that drops the trajectories that never finish. Under our pre-registered rule (fixed in writing before the data existed), it reverses for the gemma-4 model and vanishes for the other actors (§5.4).

*Prior studies of one trace corpus, and critique uptake.* Three studies report on one multi-agent trace corpus on the Omni-MATH benchmark [26, 44, 45]. The closest study separates reviewer *precision* from critique *uptake* [44]. It finds that a pipeline reviewer flags real errors more precisely than broadcast-style peer discussion does. Yet its correct critique far less often changes the next candidate, and uptake improves when the critique is placed in the solver's working context. We ask an organizational question these studies leave open: what the role structure changes beyond the division of labor, including who decides when the team asks for correction. None of them manipulates the role structure, checks that the specified division of labor is realized, or matches feedback access across conditions. Elsewhere, the failure to revise is almost always measured in a single agent, leaving no organization to ablate.

## 3 Definitions and Measurement

### 3.1 What "load-bearing" means here

An organization is specified by a tuple: agents, an interaction order, a message-passing graph, a context-visibility policy, and a *role contract* per agent. A contract asserts that an agent performs some

function (MUST decompose the problem) or that it does not (MUST NOT state a final answer). Clauses of the second kind make the specification a *division* of labor. Because a clause is prose handed to a language model, the *realized* organization is not guaranteed to match it. Nothing records which clauses were honored.

**Definition.** A component of an organization is **load-bearing** if removing or altering it reproducibly degrades task performance *under matched call and information budgets*.

The budget clause ensures we test organization rather than scale. Removing roles alongside agents, calls or context confounds the evaluation. We apply this definition *per phase*, because a team must first produce an answer and then repair it if it is wrong. The information clause covers every channel through which a signal reaches an agent. In a prompted system, feedback arrives through the same templates that carry the roles. Consequently, a role ablation can change both the roles and the feedback channel. A contrast is matched only when it changes one of them (§4). Matched access does not mean matched use: how many verdicts a team draws depends on how often it submits.

## 3.2 Measuring the realized organization

*Functional collapse (primary).* The first deliberation cycle (one pass of Planner, Executor and Reviewer) *collapses* if *all three* agents independently assert their own final answers. The pipeline then runs as three solvers in sequence rather than a division of labor. This metric targets the property the diagram asserts, not textual overlap. We measure it on the first cycle so it is not confounded with trajectory length. It is not applicable to the structurally enforced conditions because their schema forbids an answer field.

*Exact output duplication (secondary, a lower bound).* This occurs when two or more role messages are byte-identical, that is, the same raw completion character for character. It bounds collapse from below (supplement) and is never used to compare actors.

*Answer authorship, and computation.* We measure whether a Planner authors a final answer. Structural enforcement leaves the Planner no field in which to write one, so this measure serves as the manipulation check. Separately, we track whether a Planner abstains from *computing*, scored as containing no evaluated numeric equality (**P1a**, the measure of pre-registered endpoint P1; §4.1). Because P1a scores the absence of a pattern, it rewards short messages. Therefore, it cannot reliably measure a division of labor, and we use it only as a fidelity check (§4.1).

*First pass and recovery. First-pass* accuracy grades the first answer submitted to the evaluator. It need not come from the first cycle: for the role-structured team, it may follow internal deliberation without external feedback. If a trajectory never submits an answer before the budget runs out, it is graded once on its final answer, and that grade is its first pass. *Recovery* is the fraction of *failed* trajectories, those whose first graded answer was wrong, that nonetheless finish correct. Because recovery conditions on a post-treatment variable (whether the first answer failed, which the intervention can change), recovery contrasts are descriptive [32]. We also report the main contrast on unconditional final accuracy (§5.2). The verdict-matched analysis (§5.3) is conditional by construction.

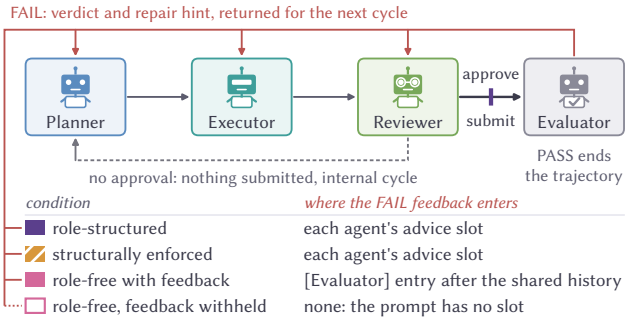



**Figure 1: One deliberation cycle and where the feedback enters. A trajectory (one problem's run, at most six cycles) advances one cycle per pass of Planner, Executor and Reviewer. The third agent's approval (the bar marks this gate) is the only route to the evaluator, which knows the reference answer and returns PASS, ending the trajectory, or FAIL with a repair hint; without approval nothing is submitted and the team deliberates internally in the next cycle. The key gives where the feedback enters per condition: the advice slot of each role prompt for the role-structured team (Planner, Executor and Reviewer prompts) and the structurally enforced one (the same roles imposed by typed message formats); for the role-free team (one generic prompt for all three agents, with no slot), an `[Evaluator]` entry in the shared history, or nowhere in the feedback-withheld diagnostic. A trajectory that never submits is graded once when the budget runs out.**

*Validation.* P1a was hand-adjudicated against a definition frozen *before* the adjudication sheet was opened. It clears a pre-set agreement threshold of 0.80 (supplement).

## 4 Intervention

We vary organization in both directions from one baseline: we remove it, and we enforce it. All conditions share three agents, one interaction order (Planner → Executor → Reviewer), the message-passing graph, context visibility (except under enforcement, below) and the total call budget. They also use the same verifier and prompt, retry budget, decoding parameters (temperature 0, that is, greedy decoding, and identical `max_tokens`), stopping rule (a PASS ends the trajectory) and grading. Every trajectory in the frozen snapshot (the pre-registered dataset, §4.2) carries the identical budget record, so the design tests organization, not scale. Besides the roles, three things differ. After a FAIL, the evaluator returns a repair hint, written with the gold answer (the reference solution) in view and instructed not to reveal it. The role-structured and enforced templates show the hint to all three agents in the next cycle, each through its role's advice slot, the template field reserved for the evaluator's feedback (Figure 1). The role-free template has no such slot, so the role-free team receives the feedback as an entry in its shared history, and a variant that withholds it serves as a diagnostic. The role-free footer (the closing instruction of the prompt) also asks every agent for an answer, and the role-structured Reviewer carries two clauses the role-free prompt lacks.

*Role-structured PER (baseline).* Role prompts are as ordinarily written. The Reviewer is told to approve only a complete answer of the right shape. In a cycle that receives a FAIL, it is told to route the work back to the Planner or Executor rather than approve.

*Role-free with feedback (main comparison).* All three agents receive one byte-identical prompt, "You are one of three agents collaboratively solving this problem. Continue the work.", followed by the problem, the shared history and the agent's slot number (its position, first to third). A footer asks for the agent's current best answer in `\boxed{}` and a closing `[Agree]` (ready to submit) or `[Continue]`, the conventions the role-structured Executor and Reviewer use. Every clause saying who does what is removed, above all the Planner's MUST NOTs on computing and answering. In every prose condition, the third agent's `[Agree]` is approval (a message with neither tag counts as `[Continue]`), and the latest boxed answer is submitted. After a FAIL, all three agents see the evaluator's verdict and repair hint, written by the same evaluator and prompt as for the role-structured team, as an `[Evaluator]` entry after the shared history; like the advice slot, it stays visible for one cycle. The two teams thus have the same access to feedback; how many verdicts each draws, and their content, depend on what it submits.

*Role-free, feedback withheld (diagnostic).* The same prompt without the `[Evaluator]` entry: nothing in this template carries the evaluator's verdict or hint. It differs from the main role-free condition only in the feedback and shows what a role ablation measures when it also removes the feedback channel.

*Structurally enforced.* The interface, not the wording, enforces specialization. The Planner emits JSON (a typed output format) against a schema with *no answer field*, so a final answer is unrepresentable. The Executor receives the parsed plan object and the problem, never the Planner's raw text, and emits a derivation and an answer. The Reviewer emits a verdict, a target span and a critique. Its `accept` promotes the Executor's answer verbatim *by reference*, because it has no field in which to *author* one. Each function stays with one agent under the same budget, but the Executor sees less context, and a rejected output can exhaust the regeneration budget (the number of times a malformed output may be regenerated). A *context-restored control* repeats this condition with the shared history restored.

*Single-agent reference.* A budget-matched single-agent protocol on the same problems guards against multi-agent gains that merely mimic a single capable agent [47]. It comes from a separate, earlier corpus (§4.2). The comparison cuts both ways, and the arms come from separate runs (supplement).

## 4.1 Instrumentation and pre-registered analyses

An intervention that never engages cannot explain an outcome, so we measure engagement. We log every rejected-and-regenerated output with its reason, count only role-boundary violations in the enforcement rate, and count answer-bearing calls per condition. One logging gap is disclosed in §5.1.

All five endpoints (the pre-specified analyses, numbered P1 to P5) were fixed in writing and committed before any intervention data existed. The full table, with the outcome for each, is in the supplement. Units are (problem, actor) pairs, and all conditions run the same problems, so every test is paired, with bootstrap intervals that resample whole problems [11]. The pre-registered primary endpoint, P4, asks whether enforcement's effect on accuracy differs by actor. It is null: no pairwise difference-in-differences (enforcement's effect in one actor minus its effect in another) survives Holm's correction for multiple comparisons [20] (smallest adjusted $p$-value, $p = 0.17$). Before correction, the 95% bootstrap interval for the Llama-3.3 model against gemma-4, +0.040 [+0.001, +0.080], only just excludes zero, and a separate resample gives $p = 0.055$. Only the gpt-oss against gemma-4 contrast had adequate statistical power for an interaction of plausible size.

The P1a manipulation objective also failed. It required enforcement to drive Planner computation-abstention to ≈1.0, but the schema forbids an *answer*, not a computation. It was re-designated to answer authorship. The role-structured versus feedback-withheld contrast was pre-registered as a secondary, exploratory replication of a published null, and splitting it into phases is post hoc. The role-free condition with feedback, and with it the main comparison, was added after the freeze (when the dataset and analysis plan were fixed). Its design and primary analysis were committed in writing before any of its data existed, but its results are exploratory, as are those of the forced-submission run (§5.3). Exclusions were fixed in advance, including the rule that decides §5.4: *unparseable trajectories count as incorrect rather than dropped.*

## 4.2 Campaign

*Task and problems.* We use the Omni-MATH benchmark of competition mathematics [14], sampled by a deterministic seeded manifest (a fixed, reproducible list of problem identifiers) so that every prefix of the run is a valid random sample. All conditions run the same problems, so every comparison is paired. A multiple-choice benchmark was ruled out before launch: on a four-option item, producing a different candidate succeeds one time in three by chance, which would inflate recovery.

*Actors and scale.* Three actor models are served from one pinned vLLM server [28]: gpt-oss-120b [34], gemma-4-31B [16] and Llama-3.3-70B [18]. The pre-registered analysis reads one frozen snapshot of 11,600 trajectories, 83,320 role messages and 17,486 gradings on 1,600 problems sampled uniformly at random from the 4,181 available. The three frozen conditions run 1,600 problems per cell (one actor in one condition on one domain) for gemma-4 and gpt-oss and 400 for Llama-3.3. The context-restored control covers gemma-4 and gpt-oss at 400 each. The evaluator is gpt-oss-120b in every cell, itself one of the actors under study (§7). The single-agent comparator is *not* part of this campaign; it comes from an earlier, separately collected corpus on the same benchmark, joined by problem identifier. Power simulated from that corpus is high for the paired per-actor main effect (§4.1 for the interaction). An early campaign, stopped when outputs cut off at the token limit were counted as role-boundary rejections, was never pooled (supplement).

*The role-free condition with feedback.* It was run after the freeze (4,748 trajectories) on the same server, problems, budget and evaluator, for all three actors on Omni-MATH and for gemma-4 and gpt-oss on the SciBench benchmark [42]. Its data are manifested

**Table 1: Functional collapse on the Omni-MATH benchmark. Share of first cycles (one pass of Planner, Executor and Reviewer) in which all three agents assert their own final answer, per actor (the model playing all three agents). RS: role-structured (Planner, Executor and Reviewer prompts); RF$^-$ and RF: role-free (one generic prompt for all three agents) with the evaluator's feedback withheld or shown. *Adjudicated no division*: the share of a judged sample of first cycles in which a model judge with a fixed rubric finds no division of labor; the judge did not see the grade but could infer the condition. The byte-duplication lower bound, the enforced condition and a sensitivity analysis over answer-marker definitions are in the supplement.**

| | functional collapse | | | adjudicated no division | |
|---|---|---|---|---|---|
| actor | RS | RF$^-$ | RF | RS | RF$^-$ |
| gemma-4-31B | 0.000 | **0.976** | **0.974** | 0.000 | **0.857** |
| gpt-oss-120b | 0.120 | **0.846** | **0.883** | 0.000 | **1.000** |
| Llama-3.3-70B | 0.000 | **0.872** | **0.900** | 0.000 | **1.000** |

separately and never pooled into a frozen cell. The two gemma-4 cells were run a second time, independently. A re-run (the first of two; supplement) of gemma-4's role-structured condition on 400 frozen problems shows no detectable drift in final accuracy (0.818 against 0.825 frozen, paired $p = 0.75$; recovery 0.198 against 0.146). Decoding at temperature 0 is not deterministic on a shared server. The two role-free conditions are byte-identical until the first verdict, yet they agree on gemma-4's first verdict for 1,496 of 1,598 problems, and their first-cycle collapse differs by up to 0.04 (Table 1). We therefore treat a cell as one sample and give the second run, where we have one, in the supplement. Prompts, runner, analysis scripts and data manifests are in the supplementary material.

## 5 Results

All statistical contrasts are paired at the problem level, with bootstrap intervals clustered by problem. Trajectories that never produce a graded answer count as incorrect, except in the first-pass contrast of §5.1. Significance is the exact McNemar test [31] on discordant pairs (problems that exactly one condition gets right).

### 5.1 R1. Roles create specialization, but not a better first answer

We delete the role clauses and keep the agents, interaction order, budget and evaluator; the role-free footer also asks every agent for its current best answer. Functional collapse, the share of first cycles in which every agent asserts its own final answer, rises from at most 0.120 with roles to at least 0.846 without them for all three actors on Omni-MATH, the gemma-4, gpt-oss and Llama-3.3 models (Table 1). Most of the change comes from the Planner (per-role rates in the supplement). The Executor answers in both conditions; the Planner seldom answers with roles (at most 0.183) and nearly always does without them, so the division of labor rests on one prohibition, its MUST NOT clause. Because the role-free footer asks for an answer, the absolute level partly reflects that request; the claim concerns the

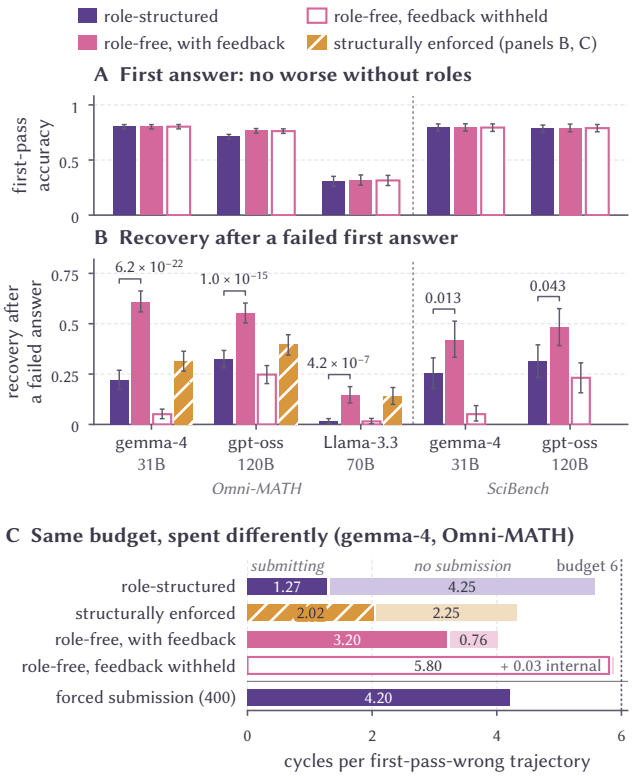



**Figure 2: Roles organize the team; the Reviewer's gate limits feedback and recovery. Conditions: role-structured (Planner, Executor and Reviewer prompts); role-free (one generic prompt for all three agents) with the evaluator's feedback shown, or withheld as a diagnostic (hollow); structurally enforced (roles imposed by typed message formats; Omni-MATH only). Actors are the models playing all three agents. (A) First-pass accuracy: whether a team's first submitted answer is correct. (B) Recovery: the share of first-pass-wrong trajectories (runs whose first graded answer was wrong) that finish correct, each condition over its own set; brackets give the exact McNemar $p$ for role-structured against role-free with feedback, paired on problems both first answer wrongly. (C) How gemma-4's first-pass-wrong Omni-MATH trajectories spend the six-cycle budget: cycles ending in a submission (the third agent approved) or internal cycles without submission (it declined). The bottom row is the exploratory forced-submission run on its own 400-problem subset: role prompts kept, the answer submitted in every cycle. (A) is over trajectories with a first verdict; in (B) trajectories that never finish count as unrecovered; error bars are bootstrap 95% intervals.**

*change*. The ordering holds for every actor under all three answer-marker definitions (the cues that count as asserting an answer; supplement).

*Content duplication and a second opinion.* For the gemma-4 and Llama-3.3 models, the role-free Planners write full solutions before

boxing an answer, and their messages share far more numbers with the Executor's than with roles, so the Executor redoes the Planner's work (supplement). A frontier model judging a sample of first cycles against a frozen rubric agrees with the collapse measure (Table 1); the judge was outcome-blind but not condition-blind (supplement), and the collapse claim does not rest on it.

*First-pass accuracy does not follow.* The collapse costs no accuracy: on problems graded both with roles and without them (the two role-free conditions are identical until the first verdict), first-pass accuracy is *identical* for gemma-4 (+0.0000, $[-0.0156, +0.0150]$), flat for Llama-3.3 and higher without roles for gpt-oss (+0.0503, $[+0.0339, +0.0666]$; Figure 2A). Because one model plays every agent, a Planner that solves the problem brings no competence the Executor lacks. The gpt-oss model may gain because its answers vary between derivations (§5.3), so extra independent solutions add chances to be right; we did not test this.

*A logging defect.* Our runner recorded a response's hidden reasoning trace only when its visible content was empty, so most of gpt-oss-120b's role-free derivations (75.7% of messages) are missing from the record (supplement). Agents pass content only, in every condition, so the defect is in the record alone and leaves no point estimate of gpt-oss's duplication. It does not manufacture the collapse: in first cycles with no affected message (106 without roles), role-free collapse is 0.821 and role-structured 0.118.

## 5.2 R2. With feedback available to both teams, the role-free team recovers more

*What a cycle is.* The pipeline runs a fixed grid of six deliberation cycles, three outer rounds of two inner rounds, each one pass of Planner, Executor and Reviewer; Figure 1 shows how a cycle ends in a *submission* or as an *internal cycle* without submission.

*Recovery and final accuracy.* Both teams can obtain the evaluator's feedback, so this is the main, exploratory comparison (Table 2; §4.1). The role-free team recovers more in all five cells. For gemma-4 on Omni-MATH it recovers 0.609 of its failed trajectories against 0.222 (each team over its own; Figure 2B), and of the 236 problems both first answer wrongly, 89 are recovered only without roles and 4 only with them ($p = 6.2 \times 10^{-22}$). Both teams reach the same first-pass rate and never lose a correct first answer, so the extra recoveries carry straight into final accuracy (0.921 against 0.846). They are genuine revisions: nearly every passing answer differs from the one first rejected, and identical evaluator prompts almost never disagree (supplement). The other two actors repeat the ordering, and the feedback leaves their first-pass accuracy essentially unchanged, as expected, since the two role-free conditions are identical until the first verdict.

*Leakage and reproducibility.* A hint written with the gold answer in view could hand it over. Hints contain it verbatim in at most 2.3% of failing verdicts in any cell, and dropping the gemma-4 trajectories on Omni-MATH that saw one barely moves either recovery rate (supplement); we did not audit partial leakage. A second, independent run of the two gemma-4 cells with feedback is close to the first, though the paired test on the SciBench benchmark weakens from $p = 0.013$ to 0.052; we report the first run throughout (supplement).

## 5.3 R3. The Reviewer's approval governs feedback opportunities

Each submission returns a verdict and, on a FAIL, a hint written with the gold answer in view, so every submission is another chance to revise. The two teams differ in how often they seize this opportunity.

*A gate on submission.* The role-structured Reviewer dictates when the team submits. For the gemma-4 model, this team submits 1.27 times per failed trajectory, compared to 3.20 for the role-free team (Figure 2C), and it submits less often in every cell (Table 2). Much of this withholding occurs before any feedback exists: 53% of the Reviewer's declines happen before the first submission, and 84 of the 316 failed trajectories never submit, so they are graded once at the end and never see external feedback. The prompt clause forbidding approval right after a FAIL covers only 19% of the declines. In later cycles where approval is allowed, the Reviewer still declines 75% of the time, against 4% for the role-free team's third agent. On Omni-MATH the clause covers at most 37% of the declines for any actor, so there the gate reflects the Reviewer's disposition rather than the letter of its prompt, plausibly because an agent instructed to critique will search for flaws; on SciBench the clause covers more (supplement).

*Removing the gate.* If this gate causes the gap, bypassing it with the role prompts unchanged should eliminate the gap. In a direct, exploratory test, we kept the role-structured prompts byte for byte but submitted the team's answer in every cycle, overriding the Reviewer. We ran it once, for gemma-4 on the 400 Omni-MATH problems of the post-freeze re-run (supplement). Every cycle the team uses then becomes a submission (Figure 2C, bottom row), and recovery surges from 0.198 to 0.613 (paired, 26 against 2, $p = 3.0 \times 10^{-6}$), matching the role-free team's 0.549 on the same problems (7 against 7). First-pass accuracy does not move, so the Reviewer's early declines did not buy a better first answer either. With the role prompts unchanged, overriding the Reviewer's approval closed the gap in this run.

*Matched on verdicts (post hoc).* A descriptive analysis examines whether the gap persists at an equal number of verdicts. On problems both teams first answer wrongly, we truncate each role-free trajectory to the number of verdicts the role-structured team received. No cell then shows a significant difference (Table 2; Llama-3.3's cell has only five recoveries in total); pooled and stratified by cell, the difference is +0.009 in favor of roles (95% interval −0.014 to +0.033). The matching is coarse: a trajectory cut off after a single FAIL cannot recover, so wherever the role-structured team received one verdict the pair ties at zero by construction, as in most of gemma-4's pairs. Restricted to problems with at least two verdicts, the difference remains not significant; recoding the scoring defect of §7 gives +0.029 (+0.004 to +0.054), a small significant margin for roles (further variants in the supplement). No variant puts the role-free team significantly ahead. A truncation cannot show that the two teams use each verdict equally well, but combined with the forced-submission run it indicates that the gap stems mainly

**Table 2: Role-structured against role-free with feedback, in every cell (one actor, the model playing all three agents, on one benchmark; $n$ problems). RS: role-structured (Planner, Executor and Reviewer prompts); RF: role-free (one generic prompt for all three agents) with the evaluator's feedback shown; RF$^-$: role-free with it withheld. A trajectory (one problem's run) is first-pass wrong when its first submitted answer is graded wrong. *Recovery*: share of first-pass-wrong trajectories that finish correct, each condition over its own set. *Recovered only by*: on the $n$ problems both RS and RF first answer wrongly, those recovered by one condition and not the other, with the exact McNemar $p$ for the two counts. *Verdict-matched* (post hoc): the same paired recovery with each RF trajectory cut off after as many evaluator verdicts (PASS or FAIL decisions) as RS received on that problem, so both get the same number of verdicts and hints. *Final accuracy*: correct at the end, over all problems, paired; never-graded trajectories count as incorrect. *Submissions*: answers the team submits (third-agent approvals) per first-pass-wrong trajectory, excluding the single end-of-budget grading. *RS never submits*: RS first-pass-wrong trajectories with no submission in six cycles, graded once at the end.**

| | recovery | | | recovered only by (paired) | | | | verdict-matched | | | final accuracy | | | submissions | | RS never |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| actor, domain ($n$) | RS | RF | RF$^-$ | $n$ | RS | RF | $p$ | RS | RF | $p$ | RS | RF | $p$ | RS | RF | submits |
| gemma-4, Omni-MATH (1,600) | 0.222 | 0.609 | 0.051 | 236 | 4 | 89 | $6.2\times10^{-22}$ | 0.165 | 0.153 | 0.69 | 0.846 | 0.921 | $1.7\times10^{-24}$ | 1.27 | 3.20 | 84/316 |
| gpt-oss, Omni-MATH (1,600) | 0.324 | 0.552 | 0.247 | 319 | 17 | 101 | $1.0\times10^{-15}$ | 0.229 | 0.216 | 0.72 | 0.803 | 0.894 | $2.7\times10^{-28}$ | 2.10 | 3.53 | 51/463 |
| Llama-3.3, Omni-MATH (400) | 0.014 | 0.147 | 0.015 | 250 | 1 | 26 | $4.2\times10^{-7}$ | 0.012 | 0.008 | 1.0 | 0.315 | 0.415 | $1.6\times10^{-7}$ | 0.32 | 2.71 | 205/276 |
| gemma-4, SciBench (574) | 0.254 | 0.419 | 0.051 | 100 | 7 | 21 | 0.013 | 0.230 | 0.190 | 0.48 | 0.847 | 0.880 | 0.0054 | 1.95 | 3.62 | 3/118 |
| gpt-oss, SciBench (574) | 0.315 | 0.483 | 0.231 | 93 | 7 | 18 | 0.043 | 0.215 | 0.247 | 0.68 | 0.852 | 0.892 | $1.9\times10^{-4}$ | 2.68 | 3.95 | 1/124 |

from how often the team requests feedback, rather than from how effectively it uses that feedback.

*The feedback-withheld diagnostic.* This diagnostic removes external feedback entirely and reveals what a role ablation measures when it also removes the feedback channel. The role-free team still submits in almost every cycle (Figure 2C), but its agents receive no signal that the answer failed, so they repeat it: for gemma-4 the team keeps a single distinct answer in 84.5% of failed trajectories. Adding the verdict and hint to the identical prompts changes this behavior. The team explores new answers, and recovery rises from 0.051 to 0.609 (paired, 139 against 2, $p = 7.2 \times 10^{-39}$). Against this diagnostic alone, roles would appear load-bearing for repair: strongly for gemma-4 (0.222 against 0.051), directionally for gpt-oss (0.324 against 0.247; $p = 0.21$), whose answer moves even without feedback (a single distinct answer in 20.8% of failed trajectories), and not for Llama-3.3 (0.014 against 0.015), whose role-structured team rarely submits.

## 5.4 R4. Generality and organizational constraints

*A second domain.* The SciBench benchmark's complete text-only slice of 574 free-response college science problems, frozen separately, tests whether the pattern belongs to the organization or to competition mathematics. The role-free prompt is byte-identical across domains; the role-structured and evaluator prompts adapt only their domain wording (supplement). For both actors run there, the gemma-4 and gpt-oss models, the ordering returns (Table 2): first-pass accuracy is flat, recovery is highest with feedback and no roles, and so is final accuracy. The recovery contrast is weaker, consistent with a looser gate: the role-structured team almost always submits at least once (all but 3 and 1 of its failed trajectories), so more of the feedback reaches it, and, as on Omni-MATH, the contrast vanishes when both teams are credited with the same number of verdicts (answer-lock rates in the supplement).

*Structural enforcement.* Enforced and role-structured prompts carry the same advice slot, so this contrast is matched on feedback access. Enforcement replaces the behavioral request with a constraint: the Planner's final answer becomes unrepresentable. The manipulation check passes: the Planner asserts a final answer in 0.000 of enforced first cycles for all three actors, against at least 0.889 without roles. Enforcement does not change what the Planner *computes*, so the adjudication still finds no functional division in 0.071 to 0.188 of enforced first cycles: an agent can be structurally stopped from answering without being stopped from doing the work. It does loosen the gate: the enforced team submits more often for every actor and, with more hints arriving, unlocks the answer for gemma-4 and Llama-3.3 (supplement).

Recovery rises most where the gate was tightest. Llama-3.3's role-structured team almost never submits, and under enforcement its recovery rises about tenfold, from 0.014 to 0.139 ($p = 3.8\times10^{-6}$), close to the role-free team's 0.147. For gemma-4 and gpt-oss the direction is the same ($p = 0.30$ and 0.11) but not resolvable against attrition: enforced trajectories that died after an incorrect first verdict count as unrecovered, as the pre-registration requires.

Improved revisability does not become end-to-end accuracy, because enforcement is brittle. A malformed or boundary-violating output must be regenerated, and when the regeneration budget runs out the trajectory dies: 8.0%, 9.9% and 17.0% of enforced trajectories never finish, against at most 2.5% in any other condition, and the weakest actor breaks the schema most often (supplement). Under the pre-registered rule, which counts these trajectories as incorrect, enforcement changes accuracy by −0.0225, −0.0050 and +0.0175 for gemma-4, gpt-oss and Llama-3.3: one significant negative and two nulls. Dropping them instead gives gains for all three actors, all significant and ordered by actor weakness, matching the direction reported for the capability equalizer [2]. That reading is survivorship bias, since it scores only the trajectories that finished. Restoring the shared history at fixed enforcement helps gemma-4, mostly through fewer trajectories dying, and not gpt-oss (supplement).

## 6 Discussion

*Roles allocate decisions as well as work.* Generating a first answer is a generation problem. Three agents reaching the same conclusion by different routes cost only tokens, so first-pass accuracy remains largely unaffected by the division of labor. After a failure, however, the team must produce something distinct from the rejected answer, and except for the gpt-oss model, teams rarely change their answer unless told it failed. Once this failure signal is provided, the role contract adds little capacity for revision: matched on the number of verdicts, the difference between the two teams nearly vanishes. The role contract primarily dictates when the team asks for external evaluation. A role-structured Reviewer withholds approval until it is satisfied, so its team draws fewer verdicts, sometimes none. In one exploratory run, overriding this Reviewer closed the gap for the gemma-4 model.

*A design trade-off, not a defect.* A Reviewer that withholds approval conserves external evaluations, which is valuable when evaluations are costly, noisy or unavailable. In our setting, with a ground-truth evaluator and a fixed cycle budget, a strict Reviewer instead limits the team's opportunities to obtain corrective feedback, and fewer verdicts mean fewer recoveries. We measure this side of the trade-off, not the full cost and performance balance. Two design implications follow for similar settings. First, if a cheap and reliable verifier is available, let the team submit frequently and gate on the verifier rather than on an internal critic. Second, whatever the role structure, route the verifier's feedback to every agent that can act on it, as the feedback-withheld diagnostic shows (§5.3).

*Information audits.* In a prompted multi-agent system, feedback, tool outputs and verifier verdicts reach the agents through slots in the same templates that carry the role instructions. Rewriting these templates during an ablation can therefore silently remove a channel, and no budget record, call count or accuracy metric reports the absence. A role ablation thus requires an information audit alongside the budget audit: the runtime inputs each template consumes must be documented, and a contrast is matched only when these input sets are identical. Even with matched access, the realized use can differ, because the organization decides how often the channels are used.

*Decision rights and information pathways.* Prior work shows that precise reviewing does not guarantee that a critique is adopted [44]. Similarly, a clean division of labor does not guarantee that a team recovers from errors. The critical factor is whether corrective information reaches the team and changes its next actions. A task-outcome evaluation alone cannot reveal this, because an aggregate score averages the stages where the organization sits idle with those where it does its work. We recommend reporting performance conditioned on a failed first attempt, measuring the realized organization instead of assuming the specified one, and treating who may submit and which agents see the feedback as core parts of the organization's specification.

## 7 Limitations

*Evaluation paradigm and oracle-gated recovery.* The recovery loop in our experiments is gated by an oracle. The evaluator triggering retries sees the ground truth, so every reported recovery is oracle-assisted, resting on a ground-truth-informed failure signal rather than autonomous self-correction. Stopping at the correct answer also hides revisions that would have turned a correct answer into a wrong one, and an evaluator that can be queried repeatedly favors the team that queries it most (§5.3).

*Scope, provenance, and experimental disclosures.* We test one PER organization, with an evaluator that is itself one of the models under study. While the ordering returns across a second domain of college science problems, the findings are limited to text-based reasoning tasks. The role-free condition with feedback, on which the main comparison rests, and the forced-submission run are exploratory follow-ups, added after the freeze and outside the pre-registration. The forced-submission run covers only one actor, one domain, and a single run, and it removes the entire gate rather than only the clause forbidding approval after a failure. Matched access to feedback does not match the realized number or content of verdicts, and we do not show that any submission policy is cost-optimal. The condition with feedback also differs in feedback presentation: role-structured agents receive feedback inside role-specific instructions, whereas role-free agents view it as a bare `[Evaluator]` entry. Finally, the context-restored control is only a *partial* control due to one-sided attrition (§5.4), and our functional measure only detects whether every agent asserted an answer, not whether their reasoning was identical.

*A scoring defect.* The automated runner graded the last *submitted* answer but recorded the last *held* answer. For the gemma-4 model on the Omni-MATH benchmark, the two disagree on 17% of role-structured first-pass-wrong trajectories, and on more in some other cells (per-cell rates in the supplement). Recoding the affected trajectories narrows the unadjusted recovery gap and shifts the pooled verdict-matched difference to +0.029 [+0.004, +0.054], which significantly favors the role-structured team (§5.3). The primary analysis keeps the scoring rule that was fixed before the data were collected.

## 8 Conclusion

Deleting multi-agent roles largely eliminated the intended division of labor in every tested model, yet first-pass accuracy did not fall. Roles do more than allocate work: they also allocate decisions. A role-structured Reviewer controls when the team submits to the external evaluator, and it frequently withholds approval. In exploratory comparisons with feedback available to both teams, the role-free team recovered more often in every tested setting. In a post hoc comparison at equal numbers of verdicts the advantage disappeared, and for gemma-4, forcing a submission in every cycle with the role prompts kept raised the role-structured team's recovery to the same level. Roles do not just assign responsibilities; they also govern decisions and information flow, which shape how a team recovers from failure. Building an effective multi-agent organization therefore goes beyond assigning roles correctly. It requires designing decision rights and information pathways that support the desired system behavior.

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